

SNEHA SATISH BHARAMBE

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PROFESSIONAL EXPERIENCE

Reliance, Jio Platforms Limited

Navi Mumbai, India

Software Development Engineer

November 2022-July 2025

- Designed, developed, and deployed **authentication and authorization REST APIs** using **Spring Boot** and **Microservices**, supporting over **5 million daily requests** while maintaining **99.9% service availability**.
- Led the migration of legacy **Spring** applications to **Spring Boot** and upgraded services from **Java 8 to Java 17**, improving application **throughput by 50%** while reducing technical debt and simplifying maintenance.
- Optimized complex SQL queries and redesigned data access patterns, **reducing query latency by 40%** while supporting event-driven processing through **Apache Kafka** for over **2 million users**.
- Built and optimized **AngularJS** front-end interfaces, improving **page load speed by 2 seconds** and reducing **user-reported issues by 10% per release cycle**.
- Streamlined **containerized** workflows using **Docker**, shortening **release cycles by 25%** and enhancing **developer output**.
- Collaborated in **Agile/Scrum teams** to deliver enterprise-grade software products on-time with less than **2% defect leakage**.

Reliance, Jio Platforms Limited

Navi Mumbai, India

Software Engineer

July 2022-October 2022

- Supervised and coached a **QA team of 25 engineers**, streamlining **test execution** and reducing **post-release defects by 30%**.
- Converted complex business requirements into **user stories and functional test cases**, enabling teams to complete sprints **2 days faster**.
- Delivered **5+ concurrent high-impact projects** with on-time, bug-free releases and 100% stakeholder approval.

Webifly Solutions

Pune, India

Web Development Intern

May 2021-January 2022

- Designed and deployed a **full-stack hospital management system** managing **500+ patient records** and **200+ staff profiles** using **HTML, CSS, and JavaScript**, reducing manual errors by **70%**.
- Engineered a **secure role-based authentication system** and backend logic in **PHP**, implementing structured **PostgreSQL database schemas** to manage patient records and ensure secure healthcare data access.
- Built and automated a billing and invoicing module with AJAX-based interactions, cutting invoice generation time by 60% for 1,000+ users and improving financial accuracy.

PROJECTS

Knowledge-Augmented CVAE for Cell Perturbation Prediction | Python, PyTorch, BioBERT

- Developed an AI-based gene perturbation prediction system using **PyTorch** and **BioBERT** on large-scale single-cell genomic datasets.
- Built the evaluation framework and benchmarked model performance using **Pearson Correlation, R², and MSE** against baseline models.
- Improved understanding of **LLMs, transformer embeddings, deep learning, and model evaluation** through applied research.

Speech Recognition System

- Developed a multilingual speech-enabled chatbot** using **Python-based Automatic Speech Recognition and Natural Language Processing** to process academic queries in Hindi and English.
- Implemented speech-to-text transcription and intent classification**, achieving **90% ASR accuracy** and enabling efficient voice-based query resolution.

SKILLS

Programming: Python, Java, JavaScript, TypeScript, C, C++, PHP, HTML, CSS

Backend & Databases: Spring Boot, AngularJS, REST APIs, Microservices, SQL, PostgreSQL, MongoDB

Cloud & DevOps: AWS Services, Docker, Linux, Git, Azure DevOps

AI & Data Technologies: Deep Learning, NLP, Speech Recognition, Apache Kafka, Distributed Systems

Testing and Documentation: Unit Testing, Integration Testing, Test Planning, Software Design Documentation, Requirements

Documentation Productivity Tools: Microsoft Excel, Microsoft Word, Microsoft PowerPoint

EDUCATION

University of Southern California

Los Angeles, California

Master's in Computer Science

August 2025-May 2027

- Coursework : Analysis of Algorithm, Database Systems, Deep Learning

Mumbai University

Navi Mumbai, India

Bachelor's in Computer Engineering | GPA:3.73/4.00

June 2018-May 2022

- Coursework: Object oriented Programming in Java, Web Designing, Machine Learning, Software Engineering, Data Structures